SENIOR HIGH SCHOOL

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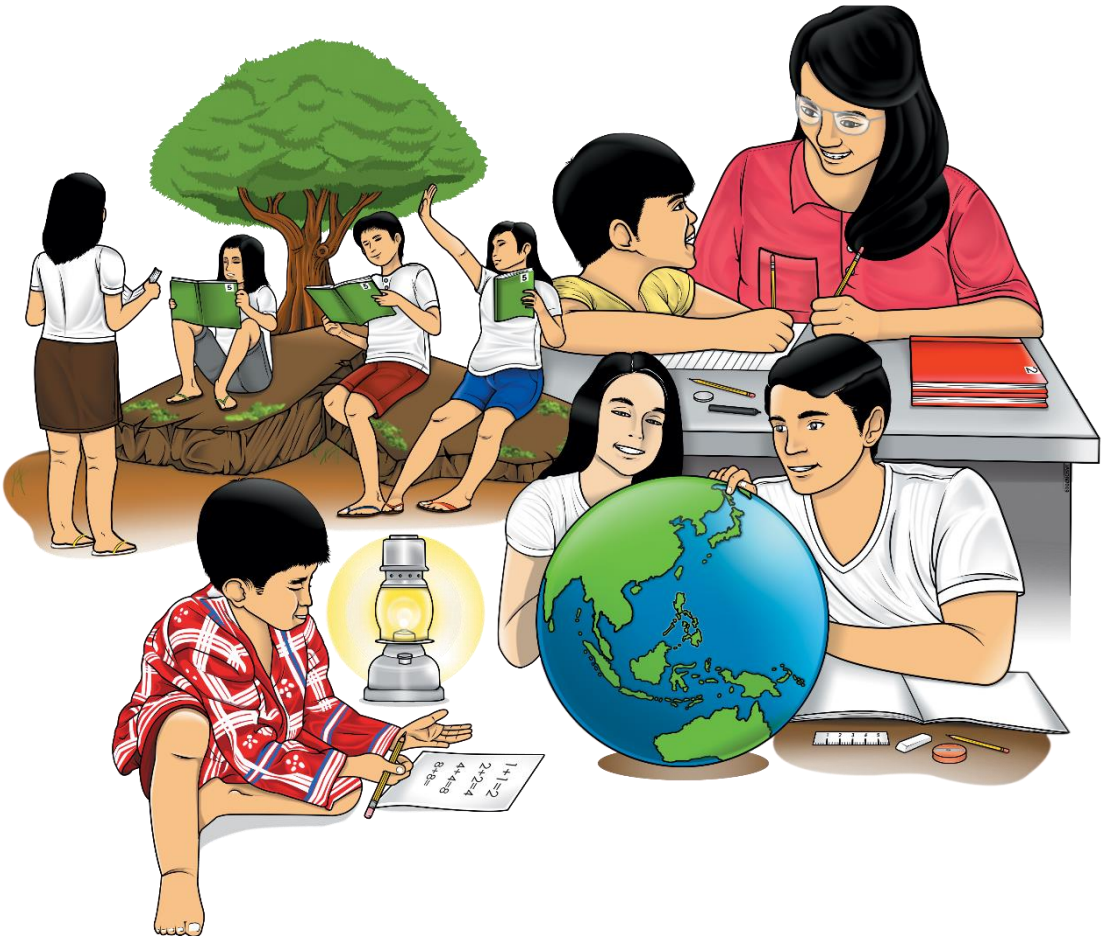
Earth Science 11

Quarter 2 – Module 2



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Quarter 2 – Module 2
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What I Need to Know

At the end of this lesson, you are expected to:

- **Describe how rocks undergo weathering. [S11ES-IIa-22]**

What I Know

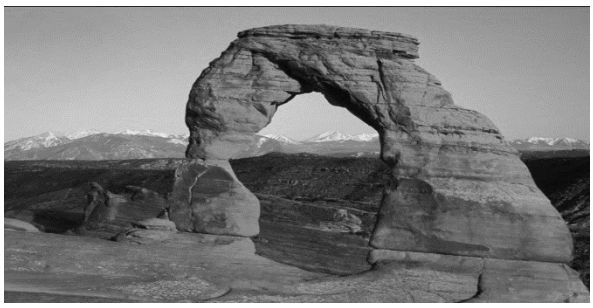
Directions: Read and analyze the questions and statements below and circle the letter of the best answer.

1. A limestone rock that contains calcium carbonate was exposed to acid rain and turns out to change its chemical composition to calcium bicarbonate. What type of weathering is present in the situation?
   A. erosion C. chemical weathering
   B. mass wasting D. physical weathering
2. The rocks and soil were carried away by running water from one place to another. What exogenic process occurs if the water serves as an agent to move the rocks and soil?
   A. erosion C. chemical weathering
   B. mass wasting D. physical weathering
3. What type of weathering do freeze, and thaw fall under?
   A. erosion C. chemical weathering
   B. mass wasting D. physical weathering
4. What do we call the sudden downslope of snow, ice, and rocks?
   A. an avalanche C. a delta
   B. an earthquake D. transpiration
5. What solid fragments of material come from the weathering of rocks?
   A. soil C. sediment
   B. mineral D. tectonic plates

Lesson 1 How weathering of rocks happens?

What's New

Directions: Observe the picture below and answer the guide questions.



Guide Questions:

1. What is the reason why the rock in the picture formed and arc?

_____

2. What geologic process is involved in the process of changing the rocks shape and composition?

_____

What is It

Weathering is the disintegration or alteration of rock in its natural or original position at or near the Earth's surface through physical, chemical, and biological processes induced or modified by wind, water, and climate. Physical weathering happens when physical forces repeatedly act on the rock. In chemical weathering, the rock disintegrates or even dissolves because a chemical reaction changes the composition of the rock. When certain types of rock come into contact with rainwater a chemical reaction occurs, slowly transforming the rock into substances that dissolve in water. As these substances dissolve, they get washed away.

What's More

Activity: Demonstrating the Weathering of Rocks

Materials:

At least four sugar cubes	Water
Used dark-colored paper	Glass
Medicine dropper	

Directions: Perform the given procedures and answer the guide questions accompanied to it.

1. Think of ways on how you can break or pulverize your rock (sugar cube).
2. Now take two new sugar cubes, and grind one against the other over a used dark-colored piece of paper.
3. Round the edges of your sugar cube.
4. Observe and record what is left in your sugar cube.
5. Take a new sugar cube and think of ways on how you can change the chemical composition of the sugar cube.
6. Use water drops to simulate rain. Place the sugar cube in a glass.
7. Fill your dropper with water and squeeze a few drops on the sugar cube. Look and feel to observe what happens.
8. Drop more water on your sugar cube.

Guide Questions:

1. Did you crush, smash, or apply force on the sugar cube? List examples of how rocks get smashed or crushed in nature.
2. What happened to the sugar cube on procedure number two? Do you see sugar dust on the used dark colored paper?
3. What do you think will happen if you drop more water on the sugar cube?
4. Can the sugar cube dissolve completely through chemical weathering? How did you say so?

What I Can Do

What is the importance of weathering in rocks?

_____

_____

Assessment

Directions: Encircle the letter of the correct answer.

1. What is the massive river of ice, rocks, and soil that moves slowly down a mountain?
   A. gorge B. canyon C. glacier D. snow
2. Which of the following contains harmful chemicals caused by burning fossil fuels?
   A. hydration rain C. acid rain
   B. oxidation rain D. ethylene
3. Which of the following breakdown rock into smaller pieces by physical means?
   A. Oxidation C. Mechanical Weathering
   B. Acid Precipitation D. Chemical Weathering
4. What happens when large amounts of soil have eroded from a steep hill?
   A. avalanche B. landslide C. sinkhole D. earthquake
5. Which of the following process where water continually seeps into cracks, freezes, and expands, eventually breaking the rock apart?
   A. abrasion B. freeze-thaw C. salt wedging D. thermal expansion

What I Need to Know

At the end of this lesson, you are expected to:

- **Explain why the Earth's interior is hot [S11ES-IIb-c-23]**

What I Know

Directions: Fill in the K-W-H-L Chart below on how heat beneath the earth generates and circulates. (5 pts.)

K	W	H	L
What do I know?	What do I want to find out?	How can I found out what I learn?	What did I learn?

Lesson 2 How does heat beneath the Earth generate and circulate?

What's New

Directions: Answer the guide questions.

Scenario: In cooking, we often boil our food to make the ingredients well cooked and safe to eat. Food boiling is necessary to us because it helps break down the food fibers for easy digestion. But how the process of boiling of food can be related to the heat movement & generation beneath the Earth?

Guide Questions:

1. What is the movement of food ingredients when boiled?

   _____

2. How can we relate this movement to heat generation beneath the Earth?

   _____

What is It

A **convection current** is a process that involves the movement of energy from one place to another. It is also called convection heat transfer. The convection currents tend to move fluid or gas particles from one place to another. These are created because of the differences occurring within the densities and temperature of a specific gas or a fluid. Convection is one among the forms of heat transfers, of which the other two are radiation and conduction. The convection process only happens in fluids. This happens due to the reason that molecules within liquids or gases are free to move.

The heat energy can be transferred by the process of convection by the difference in temperature between the two parts of the fluid. Due to this difference, the hot fluids tend to rise, whereas cold fluids tend to sink. This creates a current within the fluid called **convection current**.

What's More

Activity: **Convection Currents Activity**

Materials:

- plastic container
- food coloring
- pen
- 4 foam cups
- dropper
- white paper
- water

Procedures: Part A

1. Place a plastic box on top of four inverted foam cups. The cups may be placed on white paper for better contrast. Put a cardboard shim under each cup.
2. Half fill the plastic box with tap water. Allow the water to become still.

3. Collect a small amount of food coloring in a medicine dropper. Slowly lower the dropper into the water and release a drop of food coloring at the bottom center of the box.
4. Remove the dropper from the box slowly, make sure not to disturb the water.
5. Observe what happened to the food coloring in the water.
6. Obtain a foam cup full of hot water. Carefully slide the cup under the center of the box. Be very careful not to spill the hot water on yourself or others.
7. Collect a small amount of food coloring in the medicine dropper. After the water in the box becomes still, carefully place a drop of food coloring into the box directly over the cup of hot water.
8. Repeat steps 4 and 5.

Procedures: Part B

9. Empty the water from the box and repeat steps 1 and 2.
10. Place two cups of hot water, one at either end, under the box.
11. Place a drop of food coloring in the box directly above each cup of hot water.
12. Repeat steps 4 and 5.

Directions: Draw the path of the food coloring in each situation in the table below.

Part A - No Hot Water	Part A - With Hot Water	Part B

Guide Questions:

1. In comparison to the Earth's structure, what does each of the following represent in the model: the plastic box; the water in the box; the cup full of hot water under the box?
2. What could be used to represent lithospheric plates in the model?
3. Explain the motion of the food coloring in Part A.
4. What did you observe occurring in the region between the cups of hot water in Part B?
5. In comparison to Earth's structure, what kind of plate boundary does each of the following represent: the region directly over the cups of hot water; the region between the two cups of water?
6. Describe another example of a convection current in a natural system. How does it differ from convection currents in the asthenosphere?

What I Can Do

Directions: Draw how convection current beneath the Earth's surface and explain the process.

Assessment

Directions: Answer the given statements below and choose your answer in the word box.

_____ 1. Who is the proponent of the Tectonic Plates Theory?
_____ 2. The supercontinent that existed in the late Paleozoic and early Mesozoic Era.
_____ 3. The transfer of heat due to the bulk movement of molecules within fluids, including molten rock.
_____ 4. A hypothesis, proposed in the early 1960s, that new ocean floor is created where two plates move away from one another at mid-ocean ridges.
_____ 5. A long narrow zone where one lithospheric plate descends beneath another.

Word Bank		
Alfred Wegener	Pangea	Convection Current
Subduction Zone	Seafloor Spreading	Conduction Current

What I Need to Know

At the end of this lesson, you are expected to:

- **Describe what happens after magma is formed. [S11ES-IIc-25]**

What I Know

Directions: Circle the letter of the correct answer.

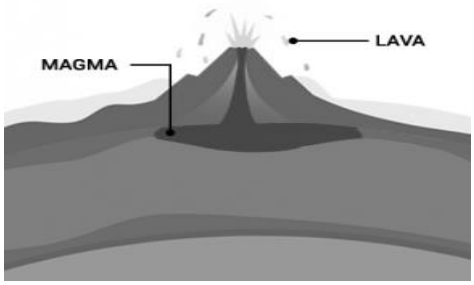
1. Intrusive igneous rocks cool slowly because they form underground. These rocks have large and sometimes interlocking crystals. Which of the following is NOT an intrusive igneous rock?
   A. Gabbro
   B. Granite
   C. Diorite
   D. Basalt
2. Extrusive igneous rocks cool quickly because they form above ground from lava that is released by volcanoes. Which of the following are extrusive igneous rocks?
   A. Basalt and Andesite
   B. Gabbro and Diorite
   C. Granite and Gabbro
   D. Andesite and Granite
3. Which mineral is the most common in a dark-colored, mafic igneous rock?
   A. Diamond
   B. Quartz
   C. Pyroxene
   D. Olivine
4. What is the origin and rate of formation of this rock?

   A. plutonic with slow cooling
   B. plutonic with rapid cooling
   C. volcanic with slow cooling
   D. volcanic with rapid cooling
5. Why are the crystals of intrusive igneous rock typically larger than those of extrusive igneous rock?
   A. because the magma cools very quickly
   B. because the magma cools much slower
   C. because of a higher water content in the magma
   D. because the temperatures of the magma are much lower

Lesson 3	What happens after magma is formed?

What's New

Directions: Observe and analyze the given illustration.



Guide Questions:

1. What will form when magma and lava cool and solidifies?
   _____
2. Will the location where magma and lava are found affect the physical structure of the materials they form? Explain.
   _____

What is It

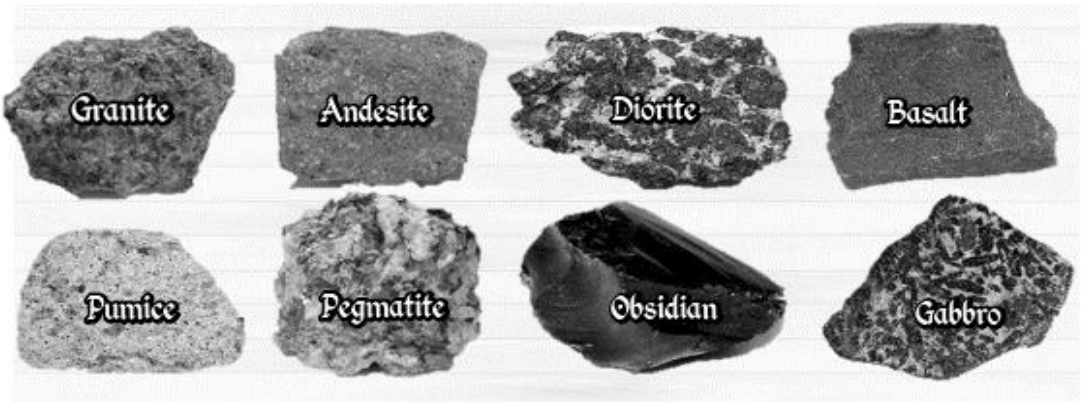
Magma is a hot, molten liquid formed due to the partial melting of rocks in the Earth's lithosphere – the outermost shell of the Earth. Differences in temperature, pressure, and structural formations in the mantle and crust cause magma to form in different ways. After its formation, magma can either experience plutonism or volcanism. Plutonism is a process by which magma rises through the crust and develops as an intrusive igneous rock beneath the Earth's surface.

What's More

Activity: Plutonism and Volcanism

Procedures:

1. Observe and analyze the given rock samples below.



2. Sort the rocks from the collection into two groups. Write the name of the rocks on their designated column.

Group A	Group B

Guide Questions:

1. What is common among the rocks in the given collection?

   _____

2. What is an igneous rock?

   _____

3. What served as your basis in classifying the rock samples into two groups?

   _____

4. Differentiate the formation of igneous rocks under plutonism and volcanism in terms of location of formation, material origin and cooling rate. Describe the factors that affect the texture and grain size of the rocks formed.

   _____

What I Can Do

1. Cite some practical uses of rocks resulting from plutonism and volcanism.

   _____

   _____

   _____

Assessment

Directions: Circle the letter of the correct answer.

1. Which is NOT a characteristic of granite, a common intrusive igneous rock?
   A. It has a phaneritic texture.
   B. It has a glassy appearance.
   C. It is generally light in terms of color.
   D. It is mostly made up of feldspar and silica.
2. Why are the crystals of an intrusive igneous rock typically larger than those of an extrusive igneous rock?
   A. Because the magma cools very quickly
   B. Because the magma cools much slower
   C. Because of a higher water content in the magma
   D. Because the temperature of the magma is much lower
3. What type of texture results when an extrusive igneous rock cools quickly that crystallization does not occur?
   A. Glassy C. Phaneritic

B. Aphanitic D. Porphyritic

4. Which is not a common extrusive igneous rock?
   A. Basalt C. Rhyolite
   B. Granite D. Andesite
5. What are the two ways igneous rocks can be classified?
   A. By luster and color C. By radioactivity and polarity
   B. By mass and volume D. By mineral proportion and grain size

What I Need to Know

At the end of this lesson, you are expected to:

- **Describe how rocks behave under different types of stress such as compression, pulling apart, and shearing. [S11ES-IId-27]**

What I Know

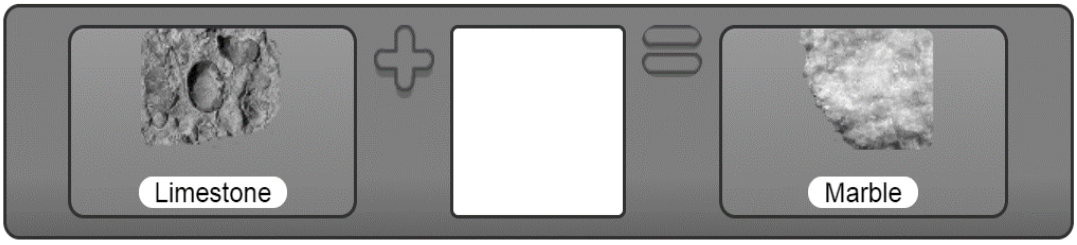
Directions: Write T if the statement is true or F if otherwise.

_____1. In Geology, stress is the force per unit area that is placed on rock.
_____2. Tension is the major type of stress at transform plate boundaries.
_____3. A rock's response to stress depends only on the type of stress.
_____4. In a normal fault, the footwall drops down relative to the hanging wall.
_____5. Confining stress always causes folds or faults.

Lesson 4 Metamorphism of Rocks

What's New

Directions: Observe and analyze the given illustration.



Guide Questions:

1. What process can turn limestone (sedimentary rock) into marble (metamorphic rock)?
   _____
2. Explain what happens during metamorphism.
   _____
   _____

What is It

Metamorphic rock is formed at the surface of the Earth through the process of metamorphism with recrystallization of minerals in rocks due to changes in pressure and temperature conditions. The three main factors/agents of metamorphism include heat, pressure, and chemically active fluids. The heat is the most important factor because it provides the energy to drive the chemical changes which results in the recrystallization of minerals. The heat increases as the depth increases.

Pressure just like heat, also increases with depth, and the buried rocks are subjected to the force or stress. Heat and pressure cause physical changes to buried rocks. Chemically active fluids enhanced the metamorphic process.

What's More

Activity 1: Complete Me!

Directions: Complete the table below by supplying the needed information.

Scheme for Metamorphic Rock Identification

TEXTURE		GRAIN SIZE	COMPOSITION	TYPE OF METAMORPHISM	COMMENTS	ROCK NAME	MAP SYMBOL
FOLIATED	MINERAL ALIGNMENT	Fine	MICA	Regional	Low-grade metamorphism of shale	Slate	
		Fine to medium	MICA, QUARTZ, FELDSPAR, AMPHIBOLE, GARNET, PYROXENE	(Heat and pressure increase with depth) ↓	Foliation surfaces shiny from microscopic mica crystals	Phyllite	
					Platy mica crystals visible from metamorphism of clay or feldspars	Schist	
	BAND-ING	Medium to coarse			High-grade metamorphism; some mica changed to feldspar; segregated by mineral type into bands	Gneiss	
NONFOLIATED		Fine	Variable	Contact (Heat)	Various rocks changed by heat from nearby magma/lava	Hornfels	
		Fine to coarse	Quartz	Regional or Contact	Metamorphism of quartz sandstone	Quartzite	
			Calcite and/or dolomite		Metamorphism of limestone or dolostone	Marble	
		Coarse	Various minerals in particles and matrix		Pebbles may be distorted or stretched	Metaconglomerate	

Rock Type	Texture (Foliated/Non-Foliated)	Composition	Rock Name	Parent	Environment (original, squeezed, or heated)

What I Can Do

Directions. Using the table, write at least 2 uses of each given example of metamorphic rock.

Rock Sample	Uses
1. Slate	
2. Marble	
3. Schist	

Assessment

Directions: Write TRUE if the statement is correct, otherwise, write FALSE.

_____1. All igneous rocks formed from magma.
_____2. Temperature is an important variable controlling the formation of magma.
_____3. The size of mineral grains in an igneous rock reveals information about the cooling history of the magma.
_____4. The slow cooling process allows the growth of large crystals.
_____5. Extrusive igneous rocks form beneath the Earth's surface.

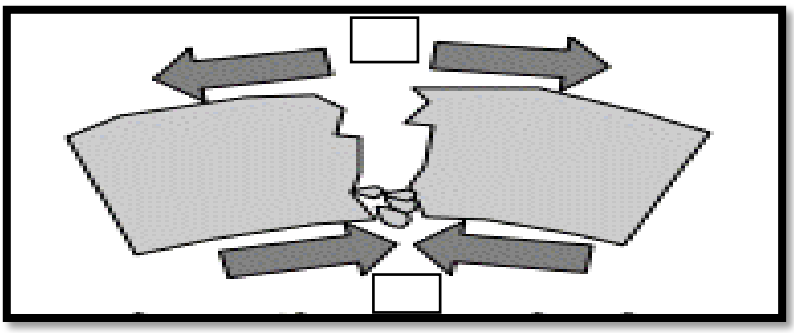
What I Need to Know

At the end of this lesson, you are expected to:

- **Describe how rocks behave under different types of stress such as compression, pulling apart, and shearing. [S11ES-IId-27]**

What I Know

Directions: Using the figure below, identify and locate the compressional and tensional stresses. Write C for compression and T for tension inside the box.



Lesson 5 Types of Stress in the Earth's Crust

What's New

While walking around in the field you spot a section of rocks in which the oldest are at top and the youngest are on the bottom. How do you explain this?

_____

What is It

The crust is broken into several parts known as the tectonic plates. When the plates are pulled or pushed together, stress occurs. Four types of stresses affect the Earth's crust: compression, tension, shear, and confining stress.

- **Compression Stress -** Compression is a type of stress that causes the rocks to push or squeeze against one another.
- **Tension Stress** - Tension is the opposite of compression. While compression forces the rocks and crust to collide and move together, tension forces the rocks to pull apart.
- **Shearing Stress** - When shear stress occurs, the force of the stress pushes some of the crust in different directions. When this happens, a large part of the crust can break off, which makes the plate size smaller. Shear stress usually happens when two plates rub against each other as they move in opposite directions.
- **Confining Stress** - When stress is applied to all sides of the crust, confining stress occurs. A deeply buried rock is pushed down by the weight of all the material above it. Since the rock cannot move, it cannot deform. This is called confining stress. When this happens, the crust compacts, which makes it look smaller.

What's More

Activity: Compression and Tension Stress Model in the Earth's Crust

Materials:

2 rectangular solid pieces of modeling clay
1 thick rubber band

A. Compression Stress

1. Simulating the two (2) rectangular solid pieces of modeling clay are your Earth's crust model for tectonic plates. Observe what happens at boundaries when tectonic plates compress against each other. Use an arrow to indicate the execution of force and draw your observation.
2. Hold one side of the one (1) rectangular solid piece of modeling clay with one hand and the other side with the other hand. Then, observe and try to push harder the other end. What would happen if this was harder like rock when it bent a certain amount?

B. Tension Stress

1. Simulating the one (1) big rectangular solid piece of modeling clay are your Earth's crust model for tectonic plates. Observe what happens at boundaries when tectonic plate pulls apart each other. Use an arrow to indicate the execution of force and draw your observation.
2. Get 1 thick rubber band and then try to stretch it. Observe what happens when you stretch the rubber band. Do they become longer or shorter?
3. This time try to exert a greater force to stretch the rubber band. Observe when the stress becomes too much on a rubber band.

Directions: Use the following space to record and explain your observations.

A. Compression Stress	B. Tension Stress

C. Shearing Stress

1. Simulating the two (2) rectangular solid pieces of modeling clay are your Earth's crust model for tectonic plates. Observe what happens at boundaries when you push some of the crust in different directions. Use an arrow to indicate the execution of force and draw your observation.

D. Confining Stress

1. Simulating one (1) big rectangular solid piece of modeling clay as your Earth's crust model for tectonic plates. Observe what happens when stress is applied to all sides of the crust. Use an arrow to indicate the execution of force and draw your observation.

Directions: Use the following space to record and explain your observations.

B. Shearing Stress	C. Confining Stress

What I Can Do

Directions: Answer the given questions and statements below.

Part A - Force Analysis

Try to break a rock of the same type of the way you broke the soap. Why is it harder to do? Use the word "force" in your answer.

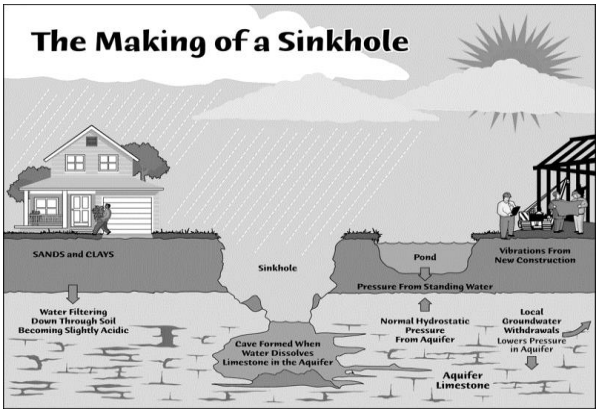
_____

True or False: Geotechnical engineers measure the amount of force it takes to break a rock using specialized and expensive equipment.

_____

Part B.

Make a simple explanation how a sinkhole formed based on the picture below.



_____
_____
_____
_____
_____
_____
_____
_____

Assessment

Directions: Circle the letter of the correct answer.

1. What do you call a stress in the earth's crust?
   A. The balancing of the force gravity on a plate with the upward force from the asthenosphere.
   B. The change in the rock layer due to a force being applied.
   C. The amount of force applied to a rock layer.
2. Which of the following refers to a compression stress?
   A. The force that squeezes and pushes rock layers together.
   B. The force that pushes parts of rock layers in opposite directions.
   C. The force that stretches rock layers apart.
3. Which of the following refers to a tension stress?
   A. The force that stretches rock layers apart.
   B. The force that pushes parts of rock layers in opposite directions.
   C. The force that squeezes and pushes rock layers together.

Directions: Complete the table by illustrating the type of stress in Earth's crust and its direction of movement.

Types of Stress in the Earth's Crust	Picture	Direction of Movement
a. Shearing		
b. Confining		

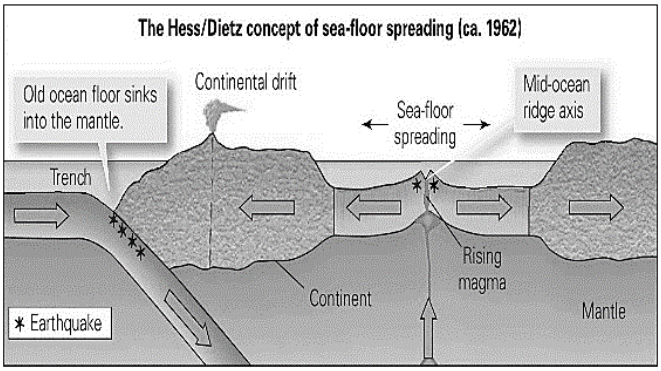
What I Need to Know

At the end of this lesson, you are expected to:

- **Explain how seafloor spreads [S11ES-IIf-32]**

What I Know

Directions: Make a simple explanation on Hess/Dietz Seafloor Spreading. (5 pts.)



_____
_____
_____
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_____

Lesson 6 How sea floor spreading happen?

What's New

You receive an e-mail offer to get rich quick by investing in an undersea gold mine on top of the Mid-Atlantic Ridge. What scientific issues should you consider before you send your money?

_____
_____

What is It

The Hess Theory

Hess argued that the **Mid-Atlantic Ridge** was a boundary where two lithospheric plates were **rifting** (being pulled apart). As that happened, rising magma from the upper part of the mantle filled in the cracks that formed in the earth's crust. After the magma solidified into **basalt** and igneous rock, additional rifting pulled those rocks apart, too. In effect, Hess proposed the existence of a magma-driven conveyor belt that continually added new seafloor, very slowly over time, widening the Atlantic Ocean basin and pushing apart the continents to either side.

What's More

Activity: Sea Floor Spreading Model

Materials: 2 sheets of 8.5" x 11" paper, ruler, scissors, colored pencils or crayons, transparent tape, masking tape

Procedure:

1. Place one sheet of binder paper so that the long side is towards you.
2. Draw a vertical line in the middle of the paper with a height of 11.5 cm leaving 5 cm on either side of the line. This line represents a mid-ocean spreading center.
3. Draw a second vertical line to the right of the center line so that it lies 3 cm from the right edge of the paper. This line represents a subduction zone.
4. Draw a third vertical line to the left of the center line so that it lies 3 cm from the left edge of the paper. This line represents another subduction zone. Label the mid-ocean ridge and subduction zones.
5. With a pair of scissors, cut the vertical lines so there will be three slits on the paper all the same height and parallel to each other. To reinforce the slits you have made, place masking tape over each one and re-cut the slit though the tape.
6. On the second sheet of paper, draw 11 bands with 2.54 cm each (1 "wide) perpendicular to the long edge of the paper.
7. Choose one color to represent normal polarity and a second to represent reversed polarity. Color alternate bands to represent periods of normal and reversed polarity. Color the band on the far left as reversed polarity.
8. Cut the paper in half parallel to the long edge to get two strips of paper as shown in Figure 2. Mark the bands on each strip with arrows to indicate alternating periods of normal (up arrow) and reversed (down arrow) polarity.
9. Insert one end of each strip of paper through the spreading center line on your first piece of paper.
10. Pull each strip of paper towards the slits nearest the margins of the paper (the subduction zones). Tape each strip to make a loop.
11. Circulate the ribbons of paper (which represent oceanic crust) to simulate the movement of ocean floor from the mid-ocean spreading center to the subduction zone. Start the movement of the ribbons with bands representing normal polarity.

What I Can Do

Reversing Poles

1. Cut six short pieces, each about 2.5 cm long, from a length of audiotape.
2. Tape one end of each piece of audiotape to a flat surface. The pieces should be Spaced 1 cm apart and lined up lengthwise in a single row.
3. Touch a bar magnet's north pole to the first piece of audiotape. Then reverse the magnet and touch its south pole to the next piece.
4. Repeat step 3 until you have applied the magnet to each piece of audiotape.
5. Sweep one end of the magnet of about 1 cm above the line of audiotape pieces. Observe what happens.
6. What characteristic of the ocean floor did you observe as you swept the magnet along the line of audiotape pieces?

Assessment

Directions: Circle the letter of the correct answer.

1. Which of the following provide evidence for seafloor spreading?
   A. Explosive stratovolcanoes and rock drilling.
   B. Floating continents and the ages of continental rocks.
   C. The pattern of seafloor ages and the pattern of seafloor magnetic

stripes.

D. The pattern of seafloor magnetic stripes and the ages of continental rocks.

2. Which of the following continents or countries is NOT bordered by the "Ring of Fire"?
   A. Asia
   B. Africa
   C. North America
   D. South America
3. What is the age of the oldest seafloor?
   A. 4 million years
   B.180 million years
   C. 1.8 billion years
   D. 4 billion years
4. Where does the deepest places on Earth located?
   A. Away from the middle of the oceans.
   B. Near the middle of the oceans.
   C. Both A and B.
   D. Neither A nor B.
5. What is the longest continuous feature on Earth?
   A. The Continents
   B. The Seafloor Spreading Centers
   C. The Ring of Fire
   D. The Atlantic Ocean

What I Need to Know

At the end of the lessons, you are expected to:

- **Explain how the movement of plates leads to the formation of folds, faults, trenches, volcanoes, rift valleys, and mountain ranges (S11ES-IIg-h-34)**

What I Know

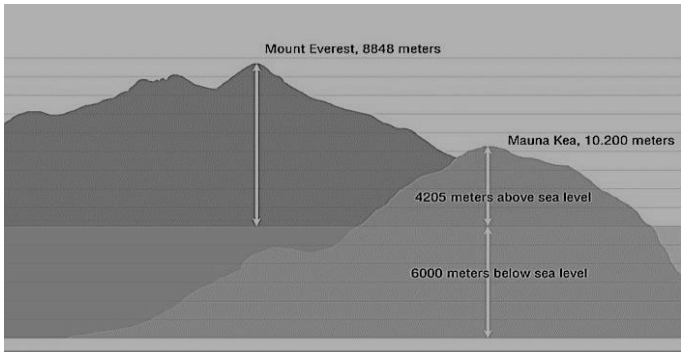
Directions: Circle the letter of the correct answer.

1. Which theory proposes that the crust and upper mantle of the Earth are composed of several large, thin, and relatively rigid plates that move relative to one another?
   A. Seafloor spreading
   B. Continental drift
   C. Bathymetry
   D. Plate tectonics
2. Which Ocean basin feature is described as the deepest feature of the seafloor?
   A. Abyssal plains
   B. Seamounts
   C. Ocean Trench
   D. Mid-oceanic ridge
3. At which boundary does plate collide with one another?
   A. Divergent B. Convergent C. Transform D. None of these
4. What type of plate boundary moves to form mid-oceanic ridges?
   A. Divergent: Continental-continental
   B. Divergent: Oceanic-oceanic
   C. Convergent: Oceanic-oceanic
   D. Convergent: Continental-continental
5. How does a transform plate boundary move?
   A. Moving away from each other
   B. Moving toward each other
   C. Sliding past each other
   D. Walking

Lesson 7	How plate tectonic processes lead to changes in Earth's surface features?

What's New

Directions: Observe the picture below and answer the guide questions.



Guide questions:

1. What is the surface feature being illustrated in the picture?

_____

2. What do you think differentiate Mount Everest from Mauna Kea?

_____

3. Which from the two is the tallest? Explain your answer.

_____

What Is It

Prior to the 1940s, very little was known about the deep oceans. During the WWII, with the advances in electronics and sonar, it became technologically possible to map the ocean floor in detail. This provided the databases to construct the first detailed maps of the important features of the ocean floor. There are three methods of measuring the ocean depths: sounding line, echo sounding and satellite altimetry. Beneath the smooth ocean surface extends an underwater landscape as complex as anything you might find on land. While the ocean has an average depth of 2.3 miles, the shape and depth of the seafloor is complex. The following are some features we might see on the seafloor: continental margin, continental shelf, continental slope, continental rise, abyssal plains, abyssal hills, mid-ocean ridges, deep-ocean trenches, seamounts, volcanic islands, island arc, guyot, submarine canyon.

What's More

Activity: **Features of Ocean Basins**

Procedures:

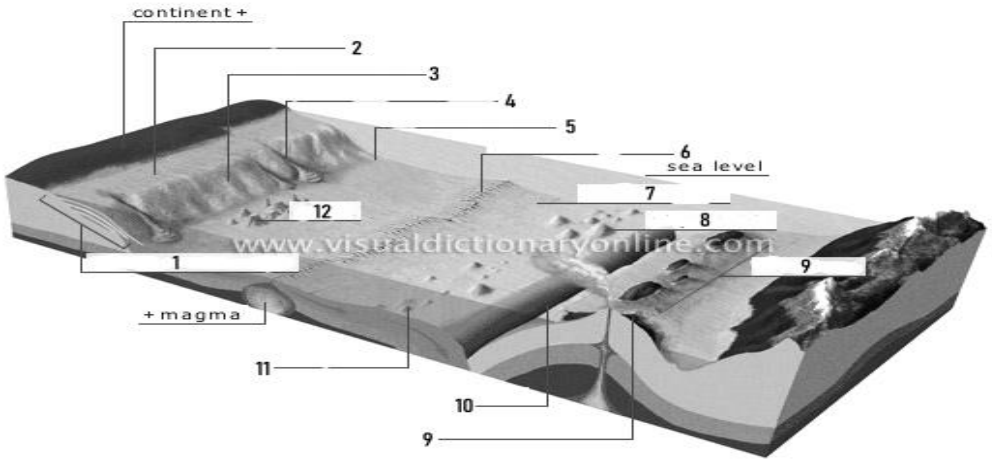
On a piece of paper, draw your own model of how seafloor would look-like. Label the surface features in your seafloor.

Guide questions:

1. Which of the features do you think is present anywhere in the actual seafloor?
   _____
2. What features are similar with land surface features?
   _____
3. How do you think they are similar?
   _____

What I Can Do

Directions: Complete the figure below by filling in the parts of ocean basin.



Assessment

Directions: Circle the letter of the correct answer.

1. This is the theory which proposes that the crust and upper mantle of the Earth are composed of several large, thin, and relatively rigid plates that move relative to one another.
   A. Seafloor spreading C. Bathymetry
   B. Continental drift D. Plate tectonics
2. Which Ocean basin feature is described as the deepest feature of the seafloor?
   A. Abyssal plains C. Ocean Trench

B. Seamounts D. Mid-oceanic ridge

3. At which boundary does plate collide with one another?
   A. Divergent B. Convergent C. Transform D. None of these
4. What type of plate boundary moves to form mid-oceanic ridges?
   A. Divergent: Continental-continental
   B. Divergent: Oceanic-oceanic
   C. Convergent: Oceanic-oceanic
   D. Convergent: Continental-continental
5. How does a transform plate boundary move?
   A. Moving toward each other C. Sliding past each other
   B. Moving away from each other D. Walking

What I Need to Know

At the end of the lessons, you are expected to:

- **Explain how the movement of plates leads to the formation of folds, faults, trenches, volcanoes, rift valleys, and mountain ranges. (S11ES-IIg-h-34)**

What I Know

Directions: Identify what is being described. Choose your answer in the word bank.

_____1. This plate boundary formed major mountain ranges of the planet.
_____2. Drives the movement of lithospheric plates.
_____3. This proposes that the crust and upper mantle of the Earth are composed of rigid plates moving above the plastic asthenosphere.
_____4. Plate movement that happens at convergent plate boundary.
_____5. Plate boundary that created the Mid Atlantic ridge.

Word Bank		
Divergent	Plate tectonics	Continental-Continental
Convergent	Lithospheric plates	Oceanic-Continental
Transform	Moves away	Convection
Oceanic-Oceanic	towards	Slides

Lesson 8 How plate tectonic move?

What's New

Directions: Read the discussion below and answer the guide question.

The Philippine Sea Plate is tectonically unusual because almost all of its boundaries are convergent. Large boundaries are the Pacific Plate on the right and the Eurasian Plate on the left. There are also shorter boundaries like Okhtosk plate to the north and the Australian plate to the south.

Guide Question:

1. What is the importance of the Philippine Sea Plate in finding evidence for some geologic link?

_____

What Is It

When talking about Earth's crust, we may think that it is like a 'cracked eggshell'. But there is more in its structure than we can ever imagine. We learned in the previous section that the ocean floor is made up of different features that somewhat similar to those in land. It is largely a result of a process called the **plate tectonics**. This theory proposes that the crust and upper mantle of the Earth are composed of several large, thin, and relatively rigid plates called tectonic plates that move relative to one another. This is the section of the lithosphere that is cracked and divided into 7 major and 8 minor plates.

What's More

Activity: Plate Tectonics

Directions: Using the table, identify what type of plate boundaries and plate movement are involved in the given geological features.

Geological Features	Plate Boundaries	Plate Movement
San Andreas Fault between the Pacific Plate and North American Plate		
Mariana Trench between Philippine Plate and the Pacific Plate		
Mid-Atlantic Ridge along the Atlantic Ocean floor		
Andes between the Nazca Plate and South American Plate		
Aleutian Islands volcanoes between Pacific plate and the North American plate		

What I Can Do

Directions: Choose a plate boundary of your choice and create a model showing the formation of geological features. Use recyclable materials in creating your model.

Rubrics for assessing models:

1. Completeness of details in the model – 35%
2. Clarity of presentation/explanation of the model – 35%
3. Creativity and resourcefulness in creating the model – 30%

Assessment

Directions: Identify what is being described. Choose your answer in the word bank.

_____1. This plate boundary formed major mountain ranges of the planet.
_____2. Drives the movement of lithospheric plates.
_____3. This proposes that the crust and upper mantle of the Earth are composed of rigid plates moving above the plastic asthenosphere.
_____4. Plate movement that happens at convergent plate boundary.
_____5. Plate boundary that created the Mid Atlantic ridge.
_____6. Plate movement that happens at transform plate boundary.
_____7. Plate boundary that creates subduction zones.
_____ 8. These are viewed like a jigsaw puzzle on top of the mantle.
_____ 9. Plate boundary that includes the Oceanic-Oceanic and Continental-Continental plates moving away from each other.
_____10. Plate movement that happens at divergent plate boundary

Word Bank

Divergent	Plate tectonics	Continental-Continental
Convergent	Lithospheric plates	Oceanic-Continental
Transform	Moves away	Convection
Oceanic-Oceanic	towards	Slides

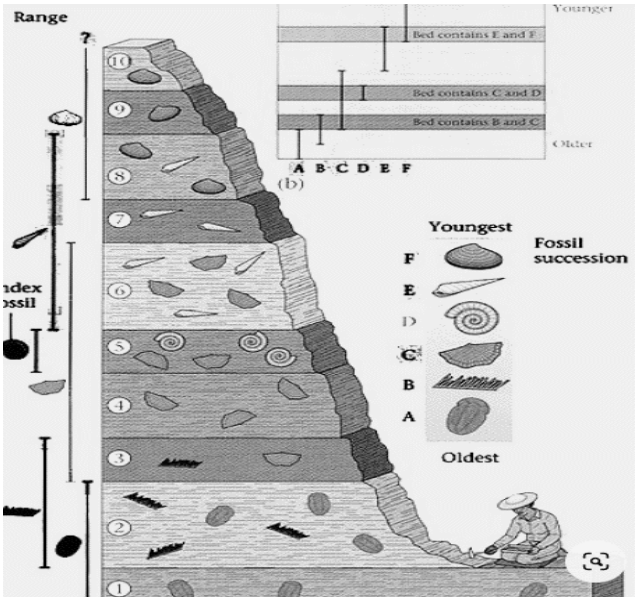
What I Need to Know

At the end of the lessons, you are expected to:

- **Describe how layers of rocks (stratified rocks) are formed [S11ES-IIh-35];**
- **Describe the different methods (relative and absolute dating) of determining the age of stratified rocks. [S11ES-IIh-i-36]**

What I Know

Directions: Analyze the figure below and answer the guide questions.



1. Numbers one to ten are rock layers, what do you think caused the layering of rocks?

   _____

   _____

2. What are found in each rock layer?

   _____

3. Order the rock layers from the OLDEST to the YOUNGEST.

   _____

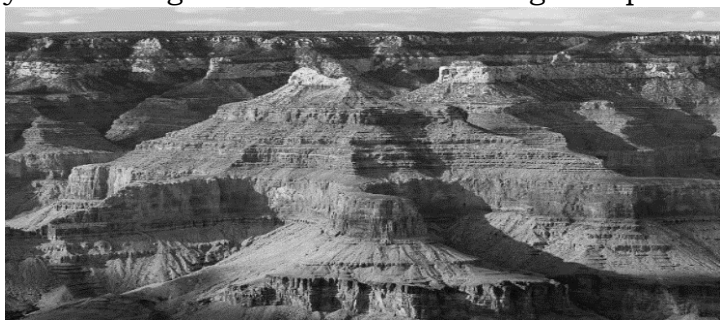
4. What are the bases for the rock layer ordering in no. 3?

   _____

Lesson 9 Formation of Stratified Rocks & Methods of Determining the Age of Stratified Rocks

What's New

Directions: Analyze the image below and answer the guide questions.



Guide Questions:

1. Were the rock layers formed at the same time or at different times? Explain your answer briefly.

   _____

   _____

2. How many years, do you think, it took for the geologic formation (Grand Canyon) to form?

   _____

What is It

Rock stratification, or **rock bedding**, is the layering of rocks which reflects the pattern of sedimentation of the original fragments of weathering. In the mid-1600's, Danish scientist Nicholas Steno studied the relative positions of sedimentary rocks. He found that solid particles settle from a fluid according to their relative weight or size. The largest, or heaviest, settle first, and the smallest, or lightest, settle last. Slight changes in particle size or composition result in the formation of layers, or beds, in the rock. Layering, or bedding, is the most remarkable feature of sedimentary rocks.

Stratigraphy is a branch of geology which deals with the arrangement and succession, origin, composition and distribution of stratified sediments or rocks on

Earth. There are two methods by which the age of stratified rocks can be determined – relative dating and absolute dating.

What's More

Activity 1: Rock Builder

Materials:

brown sugar & salt	hot water	petroleum jelly
flour	seashells	PET bottle (500 ml)
bowl		

Procedures:

1. Mix about three spoons of brown sugar with one spoon salt.
2. Smear some petroleum jelly on both sides of each three seashells (clam shells, mussel shells, etc.) using your fingers. Set the shells aside.
3. Fill a 500mL PET bottle (with its top portion cut off) with alternate layers of the different colored sands, putting the shells into different layers. Be sure to place the shells at the edge so you will be able to see them through the side of the bottle.
4. After each layer has been added, press down the topmost layer to compact the brown sugar & salt.

Guide Questions:

1. How were the colored sands set into layers?
   _____
   _____
2. What is the purpose of using the homemade plaster and pressing down on the topmost layer of the layered sand?
   _____
   _____
3. What does each sand layer represent?
   _____
4. What do the seashells represent?
   _____
5. Explain briefly how the Law of Original Horizontality is modeled in the activity._____
   _____
   _____
6. Explain briefly how the Law of Superposition is modeled in the activity.
   _____
   _____

Activity 2: Modeling Radioactive Decay

Procedures:

1. Using a timer (watch or cellphone clock), measure time of 20 seconds. After 20 seconds, carefully cut a sheet of short bond paper in half using scissors. Choose one half and set the other half aside.
2. Repeat step 1 until nine 20 s intervals have lapsed.

Guide Questions:

1. What does the whole sheet of paper used represent? _____
2. What does the half of the paper you set aside in each step represent? _____
3. How much of your "paper isotope" was left after the first 20 s interval? After the three 20 seconds intervals? After the nine 20 s intervals? Express your answers as percentages.
   _____
   _____

What I Can Do

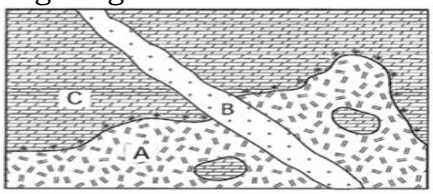
Directions: Each of the following phrases below either describes relative dating or absolute dating. On a separate sheet of paper create a Venn diagram and write the phrases that describe the two types of dating appropriately.

- Provides an estimate age of a rock or fossil
- Determines the exact age of a rock or fossil
- Compares the positions of rock layers
- Applies the stratigraphic laws
- Measures the ratio of parent radioisotope to stable daughter isotope
- Makes use of Carbon-14, Potassium-40, or Uranium-238

Assessment

Directions: Circle the letter of the correct answer.

1. Which of the following is a method that arranges geologic events based on rock sequence?
   A. absolute dating
   B. carbon dating
   C. geologic dating
   D. relative dating
2. Which of the following is a method that gives the exact age of a rock or an event in geologic history?
   A. absolute dating
   B. carbon dating
   C. relative dating
   D. stratigraphic dating
3. Which of the following stratigraphic laws states that, in any sequence of layered sedimentary rocks, the topmost layer is youngest rock layer?
   A. law of lateral continuity
   B. law of superposition
   C. law of original horizontality
   D. law of cross-cutting relationship
4. Arrange the order of the geologic features shown in the figure, from YOUNGEST to OLDEST.



A. A,B,C
B. A,C,B
C. B,A,C
D. C,A,B

5. The half-life of Cesium-137 (137Cs) is 30 years. How much 137Cs was originally present, if after 120 years 6.0 grams remained?
   A. 12.0 g
   B. 24.0 g
   C. 48.0 g
   D. 96.0 g

What I Need to Know

At the end of this lesson, you are expected to:

- **Explain how relative and absolute dating were used to determine the subdivisions of geologic time. [S11ES-IIi-37]**

What I Know

Directions: Fill in the Frayer's model below to assess your prior knowledge and understanding about dating of fossils. (5 pts.)

What is fossil dating?	How does dating of fossil happen?
Fossil Dating	
Types of fossil dating	What is the difference between? the two types of fossil dating?

Lesson 10 Use of Relative and Absolute Dating in Determining the Subdivisions of Geologic Time

What's New

Directions: Read the scenario below and answer the guide questions.

Scenario: You are about to do wash your clothes and you need to get the pile of laundry in the basket. As the week gets to its end, the pile of clothes in the laundry is getting tall.

Guide Questions:

1. Based on the pile of laundry, how are the clothes arranged in the basket?
   _____
2. How does the pile of laundry similar to rock stratification?
   _____

What is It

Geologic time scale is a record of the life forms and geologic events that have occurred during Earth's history. This scale was developed by several scientists through analysis and exploration of rock layers and fossils worldwide. Relative and absolute dating helped determine the absolute divisions in this time scale. **Relative dating** is the process of determining if one rock or geologic event is older or younger than another, without knowing their specific ages. **Absolute dating** on the other hand, is the process of determining the absolute age of an object or material.

What's More

Activity: Making Sedimentary Layers

Materials:

brown sugar and salt empty drinking glass

Procedures:

1. Place several spoonfuls of alternating brown sugar and salt in the glass to form four layers.
2. Draw the sequence and label the layers A, B, C and D (A being the bottom layer). Note that the boundary of each change in color represents a bedding plane.

Guide Questions:

1. Which layer put in first and last? _____
2. Which layer is the oldest in the sequence? _____
3. What is the overall sequence of events if there was no disturbance? (from oldest to youngest) _____

What I Can Do

1. What is the importance of relative and absolute dating in determining the subdivisions of geologic time to younger generation of learners?

_____

_____

Assessment

Directions: Write TRUE if the statement is correct, FALSE if not.

_____1. Relative dating methods are useful for establishing the absolute age of geologic events.
_____2. The Principle of Original Horizontality states that all rocks are always deposited as horizontal layers.
_____3. Radiometric dating methods used by geologists are so fraught with errors that their results are useless.
_____4. The absolute age of geologic events can never be determined.
_____5. The geologic history of Earth is continuous and without major gaps in the record.

What I Need to Know

At the end of this lesson, you are expected to:

- **Describe how index fossils (also known as guide fossils) are used to define and identify subdivisions of the geologic time scale. [S11ES-Ii-j-38]**

What I Know

Directions: Circle the letter of the correct answer.

1. What subdivision of geologic time characterized the differences in life forms?
   A. eon B. epoch C. eras D. period
2. How many percent a Precambrian takes up the history of the earth?
   A. 90% B. 80% C. 70% D. 60%
3. In which type of rocks do index fossils form?
   A. metamorphic rock B. magmatic rock
   C. sedimentary rock D. igneous rock
4. Which of the following process where a once living organism became a fossil?
   A. Index fossils B. Living fossils
   C. Fossilization D. Microfossils
5. What Era represents the "ancient life"?
   A. Proterozoic B. Paleozoic C. Mesozoic D. Cenozoic

Lesson 11 How are index fossils used to define and identify subdivisions of the geologic time scale?

What's New

Directions: Observe the picture below and answer the guide question.

Guide Question:

1. Explain how the discovery of fossils helped mankind to explore the history of the earth.

_____

_____

What is It

Index fossil, any animal or plant preserved in the rock record of the Earth that is characteristic of a particular span of geologic time or environment. A useful index fossil must be distinctive or easily recognizable, abundant, and have a wide geographic distribution and a short range through time. Index fossils are the basis for defining boundaries in the geologic time scale and f or the correlation of strata. To be considered an index fossil, it must meet 3 criteria: 1. The fossilized organism must be easily recognizable. It must be easy to identify and look unique. 2. The fossils must be geographically widespread or found over large areas so that we can use them to match layers separated by huge distances. 3. The fossil must have lived for only a short time, so that it appears in only horizontal layer of sedimentary rocks.

What's More

Activity: Fossil and Environment

Procedures:

Part 1

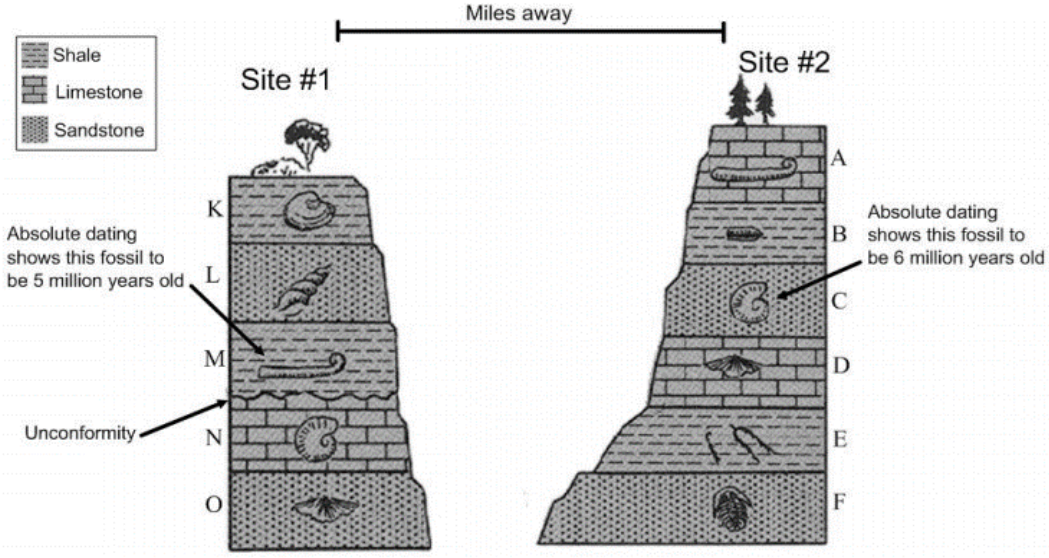
1. Observe the picture showing the cross section of rock layers.
2. Using the law of superposition, which rock layer is the oldest?
3. Using the types of fossils present in each layer, describe how the environment of this area has changed over time.

   Long ago (circle one): Land / Water

   More recent (circle one): Land / Water

Part 2

1. Draw a line connecting layers from each site that match-up by fossil type. *remember rock types may or may not match depending on environment, also, erosion can remove all or parts of layers.



Guide Questions:

1. Which is the oldest layer in Site #1 and #2? _____
2. Which layer is the oldest? _____
3. An unconformity exists between two layers on site #1. Which layer of rock on site #2 corresponds to the missing layer? _____
4. What is the approximate age of layer B, at site #2? _____
5. The wing-shaped clam fossil is found within sandstone in layer O, and within limestone in layer D. What is one possible explanation why they are not found in the same type of rock? _____

What I Can Do

1. Why index fossils are important in determining the subdivisions of the geologic time scale? _____

Assessment

Directions: Identify what is being described in the given statements. Choose your answer in the word box.

_____1. The process by which remains, or traces of living things become fossil.
_____2. The common index fossil consisting of marine artiopodan arthropods.
_____3. A fossil that belongs to the group of extinct cephalopods that exist 150 to 250 million years ago.
_____4. A preserved animal or plant in the rock record of the Earth that is used to determine the specific geologic time or environment.
_____5. A system of chronological dating that classifies geological strata in time.

Word Box

Geologic time Scale	Fossilization	Index Fossil	Crinoid
Rock Dating	Trilobite	Ammonite	

What I Need to Know

At the end of this lesson, you are expected to:

- **Describe the history of the Earth through geologic time scale. [S11ES-Iij-39]**

What I Know

Directions: Write related words or short phrases around the given central idea. Add other word/s or phrase/s around each initially written word or phrase to expand the mind map. Place your mind map in a separate sheet of paper

Lesson 12 The Geologic Time Scale and Organic Evolution Through Earth's History

What's New

Directions: Write your thoughts about what is shown in the image below.

_____
_____
_____
_____
_____
_____
_____

What is It

Based on their interpretations of the rock record, geologists have divided Earth's 4.56-billion-year history into units that represent specific amounts of time. Taken together, these time spans make up the geologic time scale. The geologic time scale is a record of the life forms and geologic events in Earth's history. The geologic time scale is divided into sections based on fossil evidence. The absolute divisions and subdivisions of the time scale, on the other hand, were determined by absolute or radiometric dating.

To see the geologic time scale you may visit:

https://www.geosociety.org/GSA/Education_Careers/Geologic_Time_Scale/GSA/timescale/home.aspx

What's More

Activity 1: A Close Look at the Geologic Time Scale

Materials:

yarn string tape measure paper
calculator marking pen masking tape

Procedures:

1. Measure out 5 meters of yarn string using a tape measure and spread it on the floor. Use masking tape to hold it down.
2. Use the following time scale:
   1 meter = 1 billion years
   10 centimeters = 100 million years
   1 centimeter = 10 million years
   1 millimeter = 1 million years
3. At the end of the string, attach a paper tag labeled "present."
4. Measure to find the spot on the string where 4.6 billion years ago would be and attach another paper tag to it labeled "Earth's Beginning."
5. Note that the timeline is broken into many time frames and are distinguished by Eras, Periods and Epochs. Calculate the length required for each and record it in table below. Then attach corresponding labeled paper tags on the string timeline.

Table 1: Length of Era or Period in the Geologic Time Scale String

Era or Period	**Million Years Ago (mya)**	**Length (m, cm or mm)**
Cenozoic Era	66 mya	
Mesozoic Era	251 mya	
Paleozoic Era	542 mya	
Precambrian	4600 mya	
Quaternary Period	1.6 mya	
Neogene Period	23 mya	
Paleogene Period	66 mya	
Cretaceous Period	146 mya	
Jurassic Period	200 mya	
Triassic Period	251 mya	
Permian Period	299 mya	
Carboniferous Period	359 mya	
Devonian Period	416 mya	
Silurian Period	444 mya	
Ordovician Period	488 mya	
Cambrian Period	542 mya	

6. Use table 2 to add word/s to the timeline of some important events in Earth's history.

Table 2. Some Important Events in Earth's History

Event Number	**Date in Years Before Present**	**Event**
1	4.6 billion	Earth formation
2	4.4 billion	Oldest mineral grain found
3	4.1 billion	Oldest piece of rock found
4	3.9 billion	Oldest evidence of a continent

5	3.8 billion	First evidence of life
6	3.5 billion	First fossils (algae and bacteria)
7	1.8 billion	Free oxygen in atmosphere
8	1.1 billion	First fossil of a complex organism (a worm)
9	540 million	First abundant life found in rock record
10	460 million	First fish
11	440 million	First land plants
12	410 million	First land animals
13	250 million	Largest mass extinction
14	247 million	First dinosaurs
15	240 million	First mammals
16	220 million	Start of Breakup of Pangaea
17	145 million	First flowering plants
18	65 million	Extinction of dinosaurs and other animals
19	30 million	Abundance of mammals and flowering plants
20	5 million	Start of Cascade Volcanic Arc
21	1.8 million	First primate in genus *Homo*
22	40,000	First *Homo sapiens*

Guide Questions:

1. What percentage of the timeline falls in the Precambrian Era? In the Phanerozoic Era? _____
2. Describe each period in terms of the key event or the first appearance of a group of organisms. (Write your answer on a separate sheet of paper.)
3. What percentage of the Earth's History has man been in existence? _____

4.1. In which two periods were there a mass extinction? _____
4.2. How many years ago did each extinction take place? _____
4.3. What species of organisms died off in each extinction? _____
4.4. What could be the cause of each mass extinction? _____

What I Can Do

Fossil record shows that species of organisms have changed over geologic time which spans millions of years. Such change through time is referred to as organic evolution. This evolution is driven by environmental changes which can affect an organism's survival.

The human species is a relatively young species (5 to 7 million years in age) compared with the other species of organisms which previously evolved and dominated in the geologic time scale.

1. Why do you think the presence of humans would influence the evolution of other species of organisms today?
   _____
   _____
2. In relation to your answer to no. 1, would the rate of evolution (time required) remain the same or would it change? Explain your answer.
   _____
   _____
   Cite one specific example to illustrate your answer to no. 2.
   _____
   _____

Assessment

Directions: Circle the letter of the correct answer.

1. What is the current age of the Earth in years?
   A. 2.5 million
   B. 16.5 million
   C. 4.6 billion
   D. 10.5 billion
2. Which subdivision covers 88% of the geologic time?
   A. Cretaceous
   B. Ordovician
   C. Paleozoic
   D. Precambrian
3. Which of the following DOES NOT belong to the group?
   A. Cenozoic
   B. Mesozoic
   C. Paleozoic
   D. Phanerozoic
4. Which Era is also known as “middle life” or the “age of dinosaurs”?
   A. Cenozoic
   B. Mesozoic
   C. Neolithic
   D. Precambrian
5. In which period did the first human appear?
   A. Cenozoic
   B. Miocene
   C. Quaternary
   D. Silurian

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